Problem H4

Møller Scattering: $e^-e^- \to e^-e^-$

Griffiths Problem 7.37 — A First QED Cross Section Calculation

Teaching version: every step explained from scratch

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§1 What Is This Problem Really Asking?

H4 — The Problem (Reworded)

The physics: Two electrons fly toward each other, exchange a virtual photon, and scatter off in new directions. This is called **Møller scattering**. We work in the high-energy limit where the electron mass can be ignored ($m = 0$).

The complication: Because the two outgoing electrons are *identical particles*, there are **two Feynman diagrams** that contribute (called $\mathcal{M}_1$ and $\mathcal{M}_2$). When you square the total amplitude $\mathcal{M} = \mathcal{M}_1 - \mathcal{M}_2$, you get four terms. Two of them ($|\mathcal{M}_1|^2$ and $|\mathcal{M}_2|^2$) can be borrowed from a result we already know (electron–muon scattering).

What you must do:

1. Compute the **interference term** $\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle$ by evaluating a trace of gamma matrices using the contraction theorems.
2. Combine all terms and express the result in **Mandelstam variables** (s, t, u).
3. Write the **differential cross section** $d\sigma/dt$.

What You'll Learn By Doing This Problem

This is your first full QED cross section from scratch. The skills are:

- Reading amplitudes from Feynman diagrams
- Squaring amplitudes with multiple diagrams (including interference)
- Converting spin sums into traces (Casimir's trick)
- Evaluating traces using contraction theorems
- Translating dot products into Mandelstam variables
- Plugging into a cross section formula

Every QED problem in the course uses this same pipeline. Learn it once here, and every subsequent problem is a variation.

§2 The Physical Setup

§2.1 What's Happening Physically?

Two electrons approach each other. They interact by exchanging a virtual photon (this is how QED describes the electromagnetic force between charged particles). After the interaction, two electrons fly out.

We label the momenta:

- p_1 = incoming electron #1
- p_2 = incoming electron #2

- p_3 = outgoing electron #1
- p_4 = outgoing electron #2

Four-momentum is conserved: $p_1 + p_2 = p_3 + p_4$.

§2.2 Why Are There Two Diagrams?

The Identical Particle Issue

In **electron–muon** scattering ($e\mu \to e\mu$), there's no ambiguity: the electron comes out as the electron, the muon comes out as the muon. One diagram.
In **electron–electron** scattering ($ee \to ee$), we have a problem: which outgoing electron is which? Both are identical. Maybe the electron that entered with momentum p_1 exits with momentum p_3 — or maybe it exits with p_4. Quantum mechanics says: **both happen, and the amplitudes add** (with a relative minus sign, because electrons are fermions).
So there are two diagrams:
Diagram 1 ("direct"): Electron p_1 scatters to p_3, electron p_2 scatters to p_4. The virtual photon carries momentum $q = p_1 - p_3$.
Diagram 2 ("exchange"): Electron p_1 scatters to p_4, electron p_2 scatters to p_3. The virtual photon carries momentum $q = p_1 - p_4$.
The **minus sign** between them (Griffiths Rule 8) comes from the antisymmetry of fermion wavefunctions — the same principle that gives the Pauli exclusion principle.

§2.3 The Amplitudes

From the Feynman rules (Griffiths Eq. 7.107), the two amplitudes are:

$$\mathcal{M}_1 = -\frac{g_e^2}{(p_1 - p_3)^2} \left[\bar{u}(3)\gamma^\mu u(1)\right] \left[\bar{u}(4)\gamma_\mu u(2)\right] \tag{1}$$

$$\mathcal{M}_2 = -\frac{g_e^2}{(p_1 - p_4)^2} \left[\bar{u}(4)\gamma^\mu u(1)\right] \left[\bar{u}(3)\gamma_\mu u(2)\right] \tag{2}$$

How to Read These

Each amplitude is built from three pieces:

1. A **coupling constant**: g_e^2 (one factor of g_e at each vertex).
2. A **propagator**: $1/(p_1 - p_3)^2$ for diagram 1 — this is $1/q^2$ for the virtual photon, where q is the photon's four-momentum.
3. Two **currents**: $[\bar{u}(3)\gamma^\mu u(1)]$ is the "current" at the top vertex (electron going from p_1 to p_3), and $[\bar{u}(4)\gamma_\mu u(2)]$ is the current at the bottom vertex. The index μ is summed over (contracted), connecting the two vertices through the photon.

$\mathcal{M}_2$ is exactly the same as $\mathcal{M}_1$, but with $3 \leftrightarrow 4$ swapped everywhere (because now electron 1 goes to p_4 instead of p_3).

The total amplitude is:

$$\mathcal{M} = \mathcal{M}_1 - \mathcal{M}_2$$

§3 What Is "Spin-Averaged Amplitude Squared"?

Why We Average Over Spins

In a typical experiment, you don't control or measure the spins of the electrons. The incoming beam has electrons with random spin orientations, and your detector counts electrons regardless of their spin. So the physically relevant quantity is:

$$\langle |\mathcal{M}|^2 \rangle \equiv \frac{1}{4} \sum_{\text{all spins}} |\mathcal{M}|^2$$

The $\sum$ adds up the squared amplitude for all 16 possible spin combinations (2 spin states for each of 4 particles: $2^4 = 16$). The $1/4$ averages over the 4 initial spin states (2×2 for the two incoming electrons).
This is what Casimir's trick is designed to compute efficiently.

§4 Squaring the Amplitude: What Are the Four Terms?

Since $\mathcal{M} = \mathcal{M}_1 - \mathcal{M}_2$, squaring gives:

$$\begin{aligned} |\mathcal{M}|^2 &= (\mathcal{M}_1 - \mathcal{M}_2)(\mathcal{M}_1 - \mathcal{M}_2)^* \\ &= |\mathcal{M}_1|^2 + |\mathcal{M}_2|^2 - \mathcal{M}_1\mathcal{M}_2^* - \mathcal{M}_1^*\mathcal{M}_2 \end{aligned} \tag{3}$$

Anatomy of the Four Terms

This is exactly like $(a-b)(a-b)^* = |a|^2 + |b|^2 - ab^* - a^*b$ for complex numbers.
Term 1: $|\mathcal{M}_1|^2 = \mathcal{M}_1\mathcal{M}_1^*$ — diagram 1 squared by itself.
Term 2: $|\mathcal{M}_2|^2 = \mathcal{M}_2\mathcal{M}_2^*$ — diagram 2 squared by itself.
Terms 3 & 4: $\mathcal{M}_1\mathcal{M}_2^*$ and $\mathcal{M}_1^*\mathcal{M}_2$ — the **interference** (or "cross") terms. These exist because the two diagrams describe the same final state, so they interfere quantum mechanically.
If the particles weren't identical, there would be only one diagram and no interference.

§5 Mandelstam Variables: A Quick Tutorial

Before computing anything, we need a convenient language for dot products of four-momenta.

What Are Mandelstam Variables?

For any $2 \to 2$ scattering process $1{+}2 \to 3{+}4$, there are three Lorentz-invariant combinations of the four-momenta:

$$s \equiv (p_1 + p_2)^2 \qquad \text{(total energy squared in the CM frame)} \tag{4}$$

$$t \equiv (p_1 - p_3)^2 \qquad \text{(momentum transfer from 1 to 3)} \tag{5}$$

$$u \equiv (p_1 - p_4)^2 \qquad \text{(momentum transfer from 1 to 4)} \tag{6}$$

They are "Lorentz invariant" — the same number in every reference frame. This means if we express our answer in terms of s, t, u, it's automatically valid in any frame.

Why Bother?

Every dot product $p_i \cdot p_j$ can be written in terms of s, t, u and the masses. This is useful because:

- Your final answer becomes frame-independent.
- All the complicated kinematic information (energies, angles) is packaged into just two independent variables (s and t, say).
- The propagator denominators are simply t and u.

§5.1 Key Properties for Massless Electrons

Expand the definitions using $p_i^2 = m_i^2 = 0$ for each particle:

$$s = (p_1 + p_2)^2 = p_1^2 + 2p_1 \cdot p_2 + p_2^2 = 0 + 2p_1 \cdot p_2 + 0 \tag{7}$$

Dot Products in Terms of s, t, u (All Masses Zero)

$$\begin{aligned} p_1 \cdot p_2 &= s/2 & \qquad p_3 \cdot p_4 &= s/2 \\ p_1 \cdot p_3 &= -t/2 & \qquad p_2 \cdot p_4 &= -t/2 \\ p_1 \cdot p_4 &= -u/2 & \qquad p_2 \cdot p_3 &= -u/2 \end{aligned} \tag{8}$$

Constraint: $s + t + u = 0$, so $u = -s - t$.
Propagator denominators: $(p_1 - p_3)^2 = t$ and $(p_1 - p_4)^2 = u$.

Where the Equalities Come From

Why is $p_1 \cdot p_2 = p_3 \cdot p_4$? Because $p_1 + p_2 = p_3 + p_4$ (momentum conservation), so $(p_1 + p_2)^2 = (p_3 + p_4)^2$, which gives $2p_1 \cdot p_2 = 2p_3 \cdot p_4$ (since all $p_i^2 = 0$). The other equalities follow similarly from momentum conservation.

§6 Term 1: $\langle|\mathcal{M}_1|^2\rangle$ — Borrowed from Electron–Muon Scattering

Why Can We Borrow This?

Look at $\mathcal{M}_1$ (Eq. 1). It has the structure:

$$\mathcal{M}_1 \propto \frac{[\text{current at vertex 1}] \times [\text{current at vertex 2}]}{q^2}$$

This is *exactly* the same structure as electron–muon scattering (Griffiths Eq. 7.106) — two distinct fermion currents connected by a single photon. The fact that both fermions happen to be electrons doesn't change the algebra of squaring and tracing.
So $\langle|\mathcal{M}_1|^2\rangle$ is given by Griffiths Eq. 7.129 with masses set to zero.

§6.1 Reading Off the Result

Griffiths Eq. 7.129 (with $m = M = 0$) gives:

$$\langle|\mathcal{M}_1|^2\rangle = \frac{2g_e^4}{t^2}\left[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3)\right]$$

Why t^2 in the denominator? Because the propagator for diagram 1 is $1/(p_1 - p_3)^2 = 1/t$, and when you square the amplitude you get $1/t^2$.

Now plug in the Mandelstam translations:

$$\begin{aligned}(p_1 \cdot p_2)(p_3 \cdot p_4) &= \frac{s}{2} \cdot \frac{s}{2} = \frac{s^2}{4} \\ (p_1 \cdot p_4)(p_2 \cdot p_3) &= \frac{-u}{2} \cdot \frac{-u}{2} = \frac{u^2}{4}\end{aligned} \tag{9}$$

So:

$$\langle|\mathcal{M}_1|^2\rangle = \frac{2g_e^4}{t^2} \cdot \frac{s^2+u^2}{4} \cdot 4 = \frac{2g_e^4}{t^2}\left[\frac{s^2+u^2}{4}\right] \cdot 4$$

Wait — let's be more careful. From Eq. 7.129, the numerator has two terms that we computed above, and the denominator is $(p_1 \cdot p_3)^2$. Since $(p_1 \cdot p_3) = -t/2$, we have $(p_1 \cdot p_3)^2 = t^2/4$. Griffiths' formula has this in the denominator:

$$\langle|\mathcal{M}_1|^2\rangle = \frac{2g_e^4}{t^2/4} \cdot \frac{s^2/4 + u^2/4}{1} = \frac{8g_e^4}{t^2} \cdot \frac{s^2+u^2}{4}$$

$\langle|\mathcal{M}_1|^2\rangle$ — Direct Term

$$\boxed{\langle|\mathcal{M}_1|^2\rangle = 2g_e^4 \frac{s^2+u^2}{t^2}}$$

§7 Term 2: $\langle|\mathcal{M}_2|^2\rangle$ — By Symmetry

The Shortcut

$\mathcal{M}_2$ is identical to $\mathcal{M}_1$ with $p_3 \leftrightarrow p_4$ swapped. Looking at the Mandelstam definitions, swapping $3 \leftrightarrow 4$ exchanges $t = (p_1 - p_3)^2$ with $u = (p_1 - p_4)^2$, while $s = (p_1 + p_2)^2$ is untouched.
So: take the answer for $\langle|\mathcal{M}_1|^2\rangle$ and swap $t \leftrightarrow u$.

$\langle|\mathcal{M}_2|^2\rangle$ — Exchange Term

$$\boxed{\langle|\mathcal{M}_2|^2\rangle = 2g_e^4 \, \frac{s^2 + t^2}{u^2}}$$

§8 Terms 3 & 4: The Interference — This Is the Real Work

Why This Term Is Different

When we squared $|\mathcal{M}_1|^2$, the spin sum factored into two separate traces (one for each fermion line). That's because $\mathcal{M}_1\mathcal{M}_1^*$ involves each fermion current paired with its own conjugate. The interference term $\mathcal{M}_1\mathcal{M}_2^*$ is different: it pairs currents from *different* diagrams, which tangles the spinors from all four particles into a **single** closed chain. The spin sum will give **one big trace** instead of a product of two smaller ones.

§8.1 Step 1: Write Out $\mathcal{M}_1\mathcal{M}_2^*$ Explicitly

From Eqs. (1) and (2):

$$\mathcal{M}_1 = -\frac{g_e^2}{t}\,[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]$$

$$\mathcal{M}_2^* = -\frac{g_e^2}{u}\,\left([\bar{u}(4)\gamma^\nu u(1)][\bar{u}(3)\gamma_\nu u(2)]\right)^* \tag{10}$$

Now we need to complex-conjugate $\mathcal{M}_2$.

Complex Conjugating a Spinor Sandwich

Recall from Section 7.7 of Griffiths: for any bilinear $[\bar{u}(a)\,\Gamma\,u(b)]$, the complex conjugate is:

$$[\bar{u}(a)\,\Gamma\,u(b)]^* = \bar{u}(b)\,\bar{\Gamma}\,u(a)$$

where $\bar{\Gamma} = \gamma^0\Gamma^\dagger\gamma^0$. For $\Gamma = \gamma^\nu$, we have $\bar{\Gamma} = \gamma^\nu$ (the gamma matrices satisfy this property). So conjugation **swaps the two spinors** and reverses their order:

$$[\bar{u}(4)\gamma^\nu u(1)]^* = \bar{u}(1)\gamma^\nu u(4)$$

$$[\bar{u}(3)\gamma_\nu u(2)]^* = \bar{u}(2)\gamma_\nu u(3)$$

Putting it together:

$$\mathcal{M}_1\mathcal{M}_2^* = \frac{g_e^4}{t\,u} \underbrace{[\bar{u}(3)\gamma^\mu u(1)]}_{\text{from } \mathcal{M}_1} \underbrace{[\bar{u}(1)\gamma^\nu u(4)]}_{\text{conjugated from } \mathcal{M}_2} \underbrace{[\bar{u}(4)\gamma_\mu u(2)]}_{\text{from } \mathcal{M}_1} \underbrace{[\bar{u}(2)\gamma_\nu u(3)]}_{\text{conjugated from } \mathcal{M}_2} \tag{11}$$

§8.2 Step 2: Identify the Spinor Chain

The Closed Loop

Look at which spinor feeds into which in Eq. (11). Reading from left to right, the adjoint and non-adjoint spinors form a chain:

$$\bar{u}(3) \;\to\; u(1), \quad \bar{u}(1) \;\to\; u(4), \quad \bar{u}(4) \;\to\; u(2), \quad \bar{u}(2) \;\to\; u(3)$$

This is a closed loop: $3 \to 1 \to 4 \to 2 \to 3$.
When we sum over spins using the completeness relation $\sum_s u^{(s)}\bar{u}^{(s)} = \not{p}$ at each link, the entire chain becomes a **single trace** of gamma matrices — you walk around the loop, writing down $\not{p}$ and γ matrices as you go.

§8.3 Step 3: Apply Completeness Relations

How the Spin Sum Becomes a Trace

Each pair $u(a)_j\;\bar{u}(a)_k$ that appears when you sum over spin s_a gets replaced by $(\not{p}_a)_{jk}$ (the completeness relation, with $m = 0$). Since all four spins are summed, every spinor pair gets replaced. Following the loop $3 \to 1 \to 4 \to 2 \to 3$:

$$\sum_{\text{spins}} \mathcal{M}_1\mathcal{M}_2^* = \frac{g_e^4}{tu}\;\mathrm{Tr}\Big[\not{p}_3\,\gamma^\mu\,\not{p}_1\,\gamma^\nu\,\not{p}_4\,\gamma_\mu\,\not{p}_2\,\gamma_\nu\Big]$$

How to read this: Start at particle 3. Write $\not{p}_3$ (from the outer product $u(3)\bar{u}(3)$). Then the vertex γ^μ. Then $\not{p}_1$ (from $u(1)\bar{u}(1)$). Then γ^ν. Then $\not{p}_4$. Then γ_μ. Then $\not{p}_2$. Then γ_ν. Close the loop with a trace.

Including the spin-averaging factor of $1/4$:

$$\langle\mathcal{M}_1\mathcal{M}_2^*\rangle = \frac{g_e^4}{4\,t\,u}\;\mathrm{Tr}\Big[\not{p}_3\,\gamma^\mu\,\not{p}_1\,\gamma^\nu\,\not{p}_4\,\gamma_\mu\,\not{p}_2\,\gamma_\nu\Big]$$

This trace involves 8 gamma matrices (each $\not{p} = \gamma^\alpha p_\alpha$ is one gamma matrix contracted with a momentum). We need to evaluate it.

§8.4 Step 4: Evaluate the Trace Using Contraction Theorems

The Strategy

Expanding an 8-gamma trace directly using trace theorem 13 would be extremely tedious. Instead, we use the **contraction theorems** — these let us collapse pairs of gamma matrices that share a summed index ($\gamma^\mu \dots \gamma_\mu$).
We'll contract the μ index first, then the ν index. Each contraction reduces the number of gamma matrices inside the trace.

§8.4.1 Contraction #1: The μ Index

We need to evaluate $\gamma^\mu \, (\not{p}_1 \, \gamma^\nu \, \not{p}_4) \, \gamma_\mu$. The gamma matrices γ^μ and γ_μ are "outside", and $\not{p}_1 \, \gamma^\nu \, \not{p}_4$ is "sandwiched" between them.

This is a contraction of the form γ_μ (three γ's) γ^μ. By Griffiths Rule 9:

$$\gamma_\mu \, \gamma^\alpha \gamma^\beta \gamma^\delta \, \gamma^\mu = -2 \, \gamma^\delta \gamma^\beta \gamma^\alpha$$

What Rule 9 Does

Rule 9 says: when you contract the outer $\gamma_\mu \dots \gamma^\mu$ around three gamma matrices, the result is -2 times the three matrices **in reversed order**. In our case:

$$\gamma^\mu \, \not{p}_1 \, \gamma^\nu \, \not{p}_4 \, \gamma_\mu = -2 \, \not{p}_4 \, \gamma^\nu \, \not{p}_1$$

The three inner matrices $\not{p}_1$, γ^ν, $\not{p}_4$ get reversed to $\not{p}_4$, γ^ν, $\not{p}_1$, with a factor of -2.

The trace becomes:

$$\mathrm{Tr}[\not{p}_3 \, (-2 \, \not{p}_4 \, \gamma^\nu \, \not{p}_1) \, \not{p}_2 \, \gamma_\nu] = -2 \ \mathrm{Tr}[\not{p}_3 \, \not{p}_4 \, \gamma^\nu \, \not{p}_1 \, \not{p}_2 \, \gamma_\nu]$$

We're down from 8 gamma matrices to 6. Progress!

§8.4.2 Contraction #2: The ν Index

Now contract γ^ν with γ_ν. The object sandwiched between them is $\not{p}_1 \, \not{p}_2$ (two gamma matrices). By Griffiths Rule 8′:

$$\gamma_\nu \, \gamma^\alpha \gamma^\beta \, \gamma^\nu = 4(a \cdot b) \qquad \text{i.e.,} \qquad \gamma^\nu \, \not{p}_1 \, \not{p}_2 \, \gamma_\nu = 4(p_1 \cdot p_2)$$

What Rule 8′ Does

Rule 8′ says: contracting the outer gammas around **two** slashed quantities gives 4 times the dot product of their momenta. All the gamma matrices disappear, leaving just a number:

$$\gamma^\nu \, \not{p}_1 \, \not{p}_2 \, \gamma_\nu = 4(p_1 \cdot p_2) = 4 \cdot \frac{s}{2} = 2s$$

Substituting back:

$$-2 \ \mathrm{Tr}[\not{p}_3 \, \not{p}_4 \cdot 2s] = -4s \ \mathrm{Tr}[\not{p}_3 \, \not{p}_4]$$

§8.4.3 The Final Trace

We're down to a trace of just two gamma matrices. By the basic trace theorem:

$$\mathrm{Tr}[\not{p}_3\, \not{p}_4] = 4(p_3 \cdot p_4) = 4 \cdot \frac{s}{2} = 2s$$

So the entire trace evaluates to:

$$-4s \cdot 2s = -8s^2$$

§8.5 Step 5: Assemble the Interference Term

$$\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle = \frac{g_e^4}{4\,t\,u} \cdot (-8s^2) = \frac{-2g_e^4\, s^2}{t\,u}$$

The other cross term $\langle \mathcal{M}_1^* \mathcal{M}_2 \rangle$ is the complex conjugate of this. But our result is already real (it's just a ratio of real Mandelstam variables), so $\langle \mathcal{M}_1^* \mathcal{M}_2 \rangle = \langle \mathcal{M}_1 \mathcal{M}_2^* \rangle$.

Interference Term

$$\boxed{\langle \mathcal{M}_1 \mathcal{M}_2^* \rangle = \langle \mathcal{M}_1^* \mathcal{M}_2 \rangle = \frac{-2g_e^4\, s^2}{tu}}$$

§9 Combining Everything

$$\langle |\mathcal{M}|^2 \rangle = \langle |\mathcal{M}_1|^2 \rangle + \langle |\mathcal{M}_2|^2 \rangle - \langle \mathcal{M}_1 \mathcal{M}_2^* \rangle - \langle \mathcal{M}_1^* \mathcal{M}_2 \rangle$$

Substituting:

$$= 2g_e^4 \frac{s^2+u^2}{t^2} + 2g_e^4 \frac{s^2+t^2}{u^2} - 2 \times \left(\frac{-2g_e^4\, s^2}{tu} \right)$$

The two interference terms combine with the minus signs:

$$-\left(\frac{-2g_e^4 s^2}{tu}\right) - \left(\frac{-2g_e^4 s^2}{tu}\right) = +\frac{4g_e^4 s^2}{tu}$$

Full Spin-Averaged $|\mathcal{M}|^2$ for Møller Scattering ($m = 0$)

$$\boxed{\langle |\mathcal{M}|^2 \rangle = 2g_e^4 \left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{4s^2}{tu} \right]}$$

Understanding the Three Pieces

First term ($\sim 1/t^2$): diagram 1 squared. Dominates when $|t|$ is small, meaning small-angle forward scattering.
Second term ($\sim 1/u^2$): diagram 2 squared. Dominates when $|u|$ is small, meaning small-angle backward scattering (the "other" electron going forward).
Third term ($\sim 1/(tu)$): the interference. Note it's **positive** — the two diagrams interfere *constructively*. This makes the cross section *larger* than what you'd get by just adding the two diagrams' rates separately. Quantum interference matters!

§10 The Differential Cross Section

§10.1 The Formula

The $d\sigma/dt$ Formula

For $2 \to 2$ scattering, the Lorentz-invariant differential cross section is (Thomson Eq. 3.37):

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s\, p_i^{*2}} \langle |\mathcal{M}|^2 \rangle$$

where p_i^* is the magnitude of the initial-state momentum in the CM frame.
For massless particles: $E^* = p^*$ and $\sqrt{s} = 2E^*$, so $p_i^* = \sqrt{s}/2$ and $p_i^{*2} = s/4$.
Also, since the final state has two **identical** particles (both electrons), we divide by the statistical factor $S = 2$ (to avoid double-counting).
Combining: $\dfrac{d\sigma}{dt} = \dfrac{1}{2} \cdot \dfrac{\langle |\mathcal{M}|^2 \rangle}{16\pi s^2}$

§10.2 Substituting

$$\frac{d\sigma}{dt} = \frac{1}{2} \cdot \frac{1}{16\pi s^2} \cdot 2g_e^4 \left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{4s^2}{tu} \right]$$

Now use $g_e^2 = 4\pi\alpha$, so $g_e^4 = 16\pi^2\alpha^2$:

$$= \frac{1}{2} \cdot \frac{2 \cdot 16\pi^2\alpha^2}{16\pi s^2} [\ldots] = \frac{\pi\alpha^2}{s^2} [\ldots]$$

Final Answer: Møller Scattering Cross Section ($m = 0$)

$$\boxed{\frac{d\sigma}{dt} = \frac{\pi\alpha^2}{s^2} \left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{4s^2}{tu} \right]}$$

with $u = -s - t$, so this is a function of s and t alone.
Here $\alpha = e^2/(4\pi) \approx 1/137$ is the fine-structure constant.

§11 Recap: The Full Pipeline

The Steps of Any QED Cross Section Calculation

1. **Draw the diagrams.** Figure out how many contribute (1 for $e\mu \to e\mu$; 2 for $ee \to ee$ because of identical particles).
2. **Write the amplitudes** using the Feynman rules (vertex factors → propagators → external spinors).
3. **Square the total amplitude.** If there are N diagrams, you get N^2 terms: N "direct" ($|\mathcal{M}_i|^2$) and $N^2 - N$ "interference" ($\mathcal{M}_i \mathcal{M}_j^*$).
4. **Spin-average** using Casimir's trick:

- Direct terms: product of two traces (one per fermion line).
- Interference terms: one big trace (closed spinor loop).

5. **Evaluate the traces** using contraction theorems (Rules 6–9) and trace theorems (Rules 10–13).
6. **Translate** all dot products into Mandelstam variables.
7. **Plug into** the cross section formula ($d\sigma/dt$ or $d\sigma/d\Omega$).

That's it. Every QED scattering problem (Bhabha, Compton, pair production, etc.) follows these same steps. The only things that change are which diagrams contribute, which spinors are u vs. v, and whether there are photon polarization vectors to deal with.

Problem H4, Griffiths 7.37 (parts).
Sources: Griffiths Ch. 7, Thomson Ch. 6, Lancaster & Blundell Ch. 40.